Analogue electronics: SQA-style questions

ENGINEERING WITH MR KELLY

National 5 · Electronics & control · Class worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ answer exam-style circuit questions under time pressure
- ☐ show working for every mark
- ☐ state formulae, units and circuit rules
- ☐ explain the effect of changing a resistance

I DO — Before you start

40 marks. Show all working.

V = IR

Voltage = current × resistance (Ohm's law)

- V in volts, I in amps, R in ohms (Ω)
- Convert first: 4.7 kΩ = 4700 Ω, 30 mA = 0.03 A

	Series	Parallel
Resistance	$R_T = R_1 + R_2 + \ldots$	$1/R_T = 1/R_1 + 1/R_2 + \ldots$ (two: $R_1R_2 \div (R_1 + R_2)$)
Current	The same everywhere	Splits: $I_T = I_1 + I_2 + \ldots$
Voltage	Shared: $V_s = V_1 + V_2 + \ldots$	The same across every branch

YOU DO — Section 1: attempt all questions

Total 40 marks.

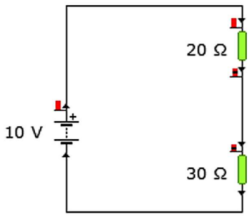
1 *A 300 Ω resistor is connected to a 12 V supply.*

a) Calculate the current through the resistor. (2 marks)

b) State the formula used. (1 mark)

c) State the unit of current. (1 mark)

2 *20 Ω and 30 Ω resistors are connected in series to a 10 V supply.*

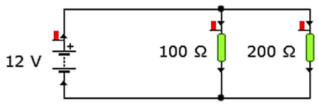


a) Calculate the total resistance. (1 mark)

b) Calculate the circuit current. (2 marks)

c) Calculate the voltage across the 30 Ω resistor. (3 marks)

3 *100 Ω and 200 Ω resistors are connected in parallel to a 12 V supply.*

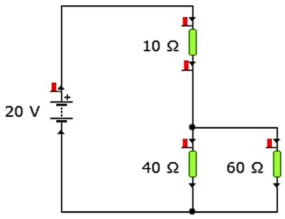


a) Calculate the total resistance. (3 marks)

b) Calculate the total circuit current. (2 marks)

c) State one characteristic of a parallel circuit. (1 mark)

4 *A 10 Ω resistor is connected in series with 40 Ω and 60 Ω resistors in parallel. The supply is 20 V.*

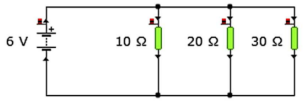


a) Calculate the resistance of the parallel section. (2 marks)

b) Calculate the total resistance. (1 mark)

c) Calculate the circuit current. (2 marks)

5 *10 Ω, 20 Ω and 30 Ω resistors are connected in parallel to a 6 V supply.*



a) Calculate the total resistance. (3 marks)

b) Calculate the current through the 10 Ω resistor. (2 marks)

6 *A student needs 50 Ω but only has two 100 Ω resistors.*

a) Draw a circuit to give 50 Ω. (2 marks)

b) Prove by calculation that the total resistance is 50 Ω. (2 marks)

7 *A 6 V supply feeds a 15 Ω resistor in series with two 30 Ω resistors in parallel.*

a) Calculate the total resistance. (2 marks)

b) Calculate the circuit current. (1 mark)

c) Calculate the voltage across the parallel section. (2 marks)

8 *A 24 V supply is connected to a 2 kΩ resistor.*

a) Calculate the current in milliamps. (3 marks)

b) The resistor is replaced by a 1 kΩ resistor. State what happens to the current and explain why. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Which “I can” statement above still needs work? Look back at the notes for that topic on the site.